Niyamic-Voice-AI-Agent.

Autonomous Enterprise Voice Agents with Warm Human Handoff

Live Demo Web App

SYSTEM ARCHITECTURE & CAPABILITIES

Autonomous Front-Line AI Voice Agents

Pre-configured multilingual agents (English, Urdu, Hindi) handling order tracking, billing exceptions, duplicate charges, and plan onboarding with deterministic policy-edge detection.

Human-in-the-Loop Supervisor Desk

Live supervisor workstation with cross-tab BroadcastChannel state sync, real-time risk scores, contextual transcript injection, single-click human takeover, and loop-return to AI.

HOW TO VERIFY & TEST THE LIVE DEMO (2-TAB WALKTHROUGH)

1. **Open Dual-Tab Evaluation Layout** — `Tab A & Tab B`

   Open **https://niyamic-voice-ai-agent.netlify.app/** in Tab B, and open **/supervisor** in Tab A side-by-side in the same browser window profile.

2. **Start Live Call Simulation** — `Tab B · /call/...`

   Click **"Start live call"** (top right). Agent Ayesha initiates greeting and confirms the customer inquiry in real-time with waveform and streaming transcript.

3. **Trigger Escalation to Human** — `Demo Controls`

   Click **Demo controls → Skip to escalation**. The conversation fast-forwards to duplicate charge order 4821. Confidence drops to **0.58**, instantly populating the supervisor queue in Tab A.

4. **Execute Warm Handoff & Return** — `Tab A · Supervisor Desk`

   In Tab A, click **"Play handoff"**. Sara Malik (human) joins the room with full context. After resolving the refund, control returns to Ayesha to close the call without customer repetition.

PRODUCTION ACTIVATION (LIVE API KEYS READY)

This web demo executes in a browser-local deterministic mock engine for instant verification without third-party API dependencies. **The codebase is already pre-architected to become fully live by simply populating your API credentials in `.env`:**

LiveKit WebRTC	Azure Speech STT/TTS	Azure OpenAI LLM	CRM / SIP Trunk
LIVEKIT_URL & KEY	AZURE_SPEECH_KEY	AZURE_OPENAI_KEY	CRM_TOKEN & SIP_URI

Live Deployment: **https://niyamic-voice-ai-agent.netlify.app/**

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